

William D. Frazer

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Earth and Planets Laboratory
Carnegie Science
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Education:

Yale University, New Haven CT, USA

Ph.D., Earth and Planetary Sciences

December 2024

M.Phil., Earth and Planetary Sciences

December 2021

Binghamton University, Binghamton NY, USA

B.S. *cum laude*, Geological Sciences: Geophysics

Spring 2019

Appointments:

Carnegie Postdoctoral Fellow

2024-

Earth and Planets Laboratory, Carnegie Science

Graduate Student Researcher

2019-2024

Department of Earth and Planetary Sciences, Yale University

Graduate Research Intern

Fall 2022

Yale Carbon Containment Lab, Yale University

Undergraduate Researcher

2016-2019

Geophysics and Remote Sensing Laboratory, Binghamton University

IRIS Intern

Summer 2018

IGPP, Scripps Institution of Oceanography, University of California San Diego

Publications:

10. Monsalve, G., Wagner, L.S., Avellaneda-Jiménez, D.S., **Frazer, W.D.**, & Alzate, M.A. 2026, “**Flat-slab subduction and overriding plate architecture beneath the northwestern Andes revealed by receiver functions.**” *Journal of Geophysical Research: Solid Earth*, 131, e2026JB034380, <https://doi.org/10.1029/2026JB034380>

9. **Frazer, W.D.**, & Park, J. 2025, “**Thick underplating and buoyancy of the Bermuda swell.**” *Geophysical Research Letters*, 52, e2025GL118279, <https://doi.org/10.1029/2025GL118279>

8. Frazer, W.D., Park, J., & Link, F. 2025, “**Crustal anisotropy as evidence for hydration of the Martian upper mantle.**” *Physics of the Earth and Planetary Interiors*, **367**, <https://doi.org/10.1016/j.pepi.2025.107434>
7. Frazer, W.D., & Park, J. 2024, “**No globally detectable seismic interfaces within Earth’s mantle transition zone: Evidence for basalt enrichment.**” *Earth and Planetary Science Letters*, **641C**, 118838, <https://doi.org/10.1016/j.epsl.2024.118838>
6. Frazer, W.D., & Park, J. 2023, “**High-resolution mid-mantle imaging with multiple-taper SS-precursor estimates.**” *Geophysical Journal International*, Volume 233, Issue 2, 1356–1371, <https://doi.org/10.1093/gji/ggac491>
5. Frazer, W.D., & Korenaga, J. 2022, “**Dynamic topography and the nature of deep thick plumes.**” *Earth and Planetary Science Letters*, **578**, 117286. <https://doi.org/10.1016/j.epsl.2021.117286>
4. Frazer, W.D., Doran, A.K., & Laske, G. 2021, “**Benchmarking Automated Rayleigh-Wave Arrival Angle Measurements for USArray Seismograms.**” *Seismological Research Letters*, **93** (2A): 763–776, <https://doi.org/10.1785/0220210189>
3. Frazer, W.D., & Park, J. 2021, “**Seismic evidence of mid-mantle water transport beneath the Yellowstone region.**” *Geophysical Research Letters*, **48**, e2021GL095838, <https://doi.org/10.1029/2021GL095838>
2. Nikulin, A., de Smet, T., Baur, J., Frazer, W.D., & Abramowitz, J. 2018, “**Detection and identification of remnant PFM-1 ‘Butterfly Mines’ with a UAV-based thermal-imaging protocol.**” *Remote Sensing*, 10(11), 1642, <https://doi.org/10.3390/rs10111672>
1. Nikulin, A., de Smet, T., Frazer, W.D., Baur, J., Abramowitz, J., Finan, D., Denara, S., Aglietti, N., & Campos, G. 2018, “**Drones and “butterflies”: A low-cost UAV system for rapid detection and identification of unconventional minefields.**” *Journal of Conventional Weapons Destruction*, 22.3.

Publications in review, revision, and prep.

11. Frazer, W.D., “**Discontiguous underplating beneath the Hawaiian Islands.**” in review at *Geophysical Research Letters*
12. Frazer, W.D., “**Seismic Detectability of Mantle Transition Zone Mineral Phase Changes.**” in prep.

Awards:

Carnegie Postdoctoral Fellowship
Earth and Planets Laboratory, Carnegie Science

2024

Prize Teaching Fellowship 2023
Graduate School of Arts and Science, Yale University
Student nominated for outstanding performance and promise as a teacher (EPS 212 Oman Trip)

Excellence in Teaching Prize 2023
Department of Earth and Planetary Sciences, Yale University
For outstanding contributions to the teaching process (EPS 212 Oman Trip)

Glenn G. Bartle Class of '52 Award 2019
Department of Geological Sciences, Binghamton University
Awarded to an outstanding senior for academic excellence and service to the department

NSF GRFP Honorable Mention 2019
National Science Foundation

Distinguished Independent Work 2019
Department of Geological Sciences, Binghamton University
Awarded for completion of advanced research as an undergraduate

Provost's Award for Excellence in Undergraduate Research 2019
Binghamton University
Awarded to an undergraduate for excellence in research and scholarship

Invited Talks:

Geological Society of Washington, Washington D.C. October 2026

Earth and Planets Laboratory, Carnegie Science January 2026

Geology and Geophysics departmental seminar, Texas A&M University February 2025

Earth Sciences departmental seminar, Binghamton University April 2023

Funding:

National Science Foundation – GEO SPSE “**An Investigation of Plume-less Intraplate Volcanism: Earthquake Detection, 3D Seismic Imaging, and Mineral Physics Beneath Bermuda.**” \$510,012, Pending

Selected First Author Conference Presentations:

Frazer, W.D., “Lithospheric remnants of magmatism beneath the Hawaiian Islands.” Talk at 2026 AGU Fall meeting (upcoming)

Frazer, W.D. & Park, J. “Thick underplating and buoyancy of the Bermuda swell.” Invited Talk at GAC-MAC 2026 (unable to attend)

Frazer, W.D. & Wagner, L., “Geophysical Constraints on the Lithosphere Beneath Plume-less Ocean Bathymetric Swells.” Presentation at 2025 AGU Fall Meeting

Frazer, W.D., “Global Observations of Melt-induced Low Velocity Zones Surrounding the Mantle Transition Zone.” Presentation at 2025 SSA Annual Meeting

Frazer, W.D. & Park, J. “Seismic Detectability of Mantle Transition Zone Mineral Phase Changes.” Presentation at 2024 AGU Fall Meeting

Frazer, W.D. & Park, J. “No Globally Detectable Seismic Interfaces within Earth’s Mantle Transition Zone: Evidence for Basalt Enrichment.” Poster at 2023 AGU Fall Meeting

Frazer, W.D. & Park, J. “Does the 520-km discontinuity exist? / A global search for water in Earth’s mantle transition zone.” Poster at 2023 Interior of the Earth Gordon Research Conference

Frazer, W.D. & Park, J. “Global High-Resolution mid-Mantle Imaging with Multiple-Taper SS-Precursor Estimates.” Poster at 2023 EGU General Assembly

Teaching:

TA, EPS 312, Structural Geology	Fall 2022
TA, EPS 210, Geology of North America	Spring 2022
TA, EPS 110, Dynamic Earth	Fall 2020
TA, G&G 100, Natural Disasters	Fall 2019

Service:

Reviewer: Nature Communications, Geophysical Journal International, Geophysical Research Letters, Journal of Geophysical Research: Solid Earth, G-Cubed, Seismica, National Science Foundation

Postdoctoral Representative

Carnegie Institution Postdoctoral Association	2026-
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Peer mentor

Yale EPS Graduate Student Peer Mentoring Program	2023-2024
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Co-convener, EGU

Physics-based earthquake modeling and engineering sessions	2023
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Dana Club Co-President

Yale EPS graduate student organization	2021-2022
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Co-representative for Geophysics

Yale EPS Colloquium Committee	2021-2022
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Field and Ship Experience:

Lead Investigator: Bermuda Earthquake and Structure Test (BEST), Bermuda (~3 weeks+)	2025-
Co-chief Scientist: R/V Marcus G. Langseth, Eastern Pacific (~1 month)	Fall 2025
MUSICA deployment recovery, Colombian Andes (~3 weeks)	Spring 2025
The Rutgers and Yale Rapid Seismic Array deployment, NJ, USA (~2 days)	Spring 2024
New England Seismic Network (NEST), MA, NH, VT, USA (~2 weeks total)	2019-2022
R/V Marcus G. Langseth, Off coast AK, USA (~3 weeks)	Summer 2019
R/V Robert Gordon Sproul, Off coast San Diego, CA, USA (~3 Days)	Summer 2018
Nantahala National Forest, NC, USA (~10 days)	Summer 2018
Poverty Point, LA, USA (~1 week)	Summer 2017

Field Trip Leadership:

Lead TA Yale EPS 212, Global Tectonics Field Trip – Sicily, Italy (~10 days)	Spring 2024
Co-Lead Yale EPS Dept. Field Trip – Mainland Greece, Cyclades, and Crete (~15 days)	Summer 2023
Lead TA Yale EPS 212, Global Tectonics Field Trip – Samail Ophiolite, Oman (~10 days)	Spring 2023
Lead TA Yale EPS 212, Global Tectonics Field Trip – Death Valley, CA, USA (~7 days)	Spring 2022
Co-Lead Yale EPS Dept. Field Trip – Western, USA (~3 weeks)	Summer 2021

Additional Training

6-week Advancing Inclusive Mentoring (AIM) program	Fall 2026
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Outreach and Media

Carnegie Science Earth and Planets Laboratory Open House	October 2026
Interview with Royal Gazette on BEST network	June 2026
@CarnegiePlanets Instagram takeover for BEST network service run Week-long content shared on Earth and Planets Laboratory Instagram	May 2026
“Bermuda: A Geologic Mystery,” Public Lecture: YouTube Recording Neighborhood Lecture Series, Earth and Planets Laboratory, Carnegie Science	May 2026
Interview with Live Science on Bermuda GRL Paper, coverage by ABC news	December 2025
Public tours of Earth and Planets Laboratory Field Geophysics Laboratory	Fall 2025

Computational Skills

Python, MATLAB, high-performance computing (Slurm), AxiSEM3D, UNIX
Familiar with Fortran, R, Generic Mapping Tools